core, datasets and scripts: the shared parts

core: turning counts into certificates

`bi_certificate.py` turns a count of correct guesses into a BI value. `baseline_metrics.py` has the other information measures (PI, HI and mutual information) used for comparison. `data_loader.py` finds and loads the trace files and can also make artificial traces. Every experiment imports these instead of carrying its own copy.

The loader has a self-test that needs no data:

```
python -m core.data_loader
```

```
$ python -m core.data_loader

Dataset Status
==================================================
  ✗ ascad: NOT FOUND
      Not found. Run: python -m datasets.setup_datasets --ascad
  ✗ aes_hd: NOT FOUND
      Not found. Run: python -m datasets.setup_datasets --aes-hd

Data Loader Module - Example Usage
==================================================

Generating synthetic data...
  Traces shape: (10000, 50)
  Labels shape: (10000,)

Creating train/val/holdout split...
  Training: 6000
  Validation: 2000
  Holdout: 2000

Creating representations...
  raw: shape = (6000, 50)
  poi_5: shape = (6000, 5)
  pca_10: shape = (6000, 10)
  window_10_30: shape = (6000, 20)
[exit code 0, 1 s]
```

datasets: where the traces live

All downloaded traces go under `datasets/raw/`, which git ignores. `setup_datasets.py` can fetch two trace sets that the other scripts do not: `--ascad` (the plain ASCAD file) and `--aes-hd`. It can also check which files are present:

```
python -m datasets.setup_datasets --info
python -m datasets.setup_datasets --verify --data-dir datasets/raw
```

```
$ python -m datasets.setup_datasets --info; python -m datasets.setup_datasets --verify --data-dir datasets/r
    aw

======================================================================
Dataset Information
======================================================================
```

```
ASCAD (Fixed Key)
----------------------------------------
  Description: ANSSI electromagnetic traces, fixed key variant
  Size: ~4700 MB
  Profiling: 50,000
  Attack: 10,000
  Features: 700

ASCAD (Variable Key)
----------------------------------------
  Description: ANSSI electromagnetic traces, variable key variant
  Size: ~420 MB

AES-HD Extended
----------------------------------------
  Description: Power traces with Hamming Distance leakage
  Size: ~500 MB
  Samples: 500,000
  Features: 1250

======================================================================
Dataset Verification
======================================================================

ASCAD: Not found at datasets/raw/ascad/ASCAD.h5

AES-HD: Not found at datasets/raw/aes_hd/aes_hd_ext.npz
[exit code 1, 1 s]
```

The check reports the two optional files as missing unless you fetched them; that is fine for all four levels.

scripts: extra tables and plots

`plot_bi_loc_radius_curves.py` draws the curves of Figure 2. `generate_paper_plots.py` draws the extra plots of the long version from the results of guides 07 to 09; a few of its inputs come from runs whose scripts are not in this repository, and `scripts/README.md` lists them. The two figures of the paper do not depend on this folder.

```
python -m scripts.plot_bi_loc_radius_curves --help
```

```
$ python -m scripts.plot_bi_loc_radius_curves --help
usage: plot_bi_loc_radius_curves.py [-h] [--csv CSV] [--out-dir OUT_DIR]
                                    [--audited-points-csv AUDITED_POINTS_CSV]
                                    [--point-label POINT_LABEL]
                                    [--output-stem OUTPUT_STEM]

options:
  -h, --help            show this help message and exit
  --csv CSV
  --out-dir OUT_DIR
  --audited-points-csv AUDITED_POINTS_CSV
                        Optional CSV of concrete audited radius points to
                        overlay with star markers.
  --point-label POINT_LABEL
                        Legend text for the star markers.
  --output-stem OUTPUT_STEM
                        File name (without extension) of the figure.
[exit code 0, 1 s]
```

docs

`docs/EXPERIMENT_MAP.md` lists every table and figure of the paper and the script that makes it.